Turn algebra problems into geometry! I usually do the reverse....

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1 Formula sheet

These formulas should be useful for the following problems.

- Pythagorean theorem:

$$a^2 + b^2 = c^2 \quad \text{or} \quad \sqrt{a^2+b^2} = c \quad \text{or} \quad \sqrt{c^2-a^2} = b.$$

- Law of cosines: $c^2 = a^2 + b^2 - 2ab\cos C$.
- Heron's formula:

$$\begin{aligned} [ABC] &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \frac{1}{4}\sqrt{(a+b+c)(a+b-c)(a-b+c)(-a+b+c)} \\ &= \frac{1}{4}\sqrt{(a^2+b^2+c^2)^2 - 2(a^4+b^4+c^4)}. \end{aligned}$$

- $[ABC] = \frac{1}{2}ab\sin C$.
- Stewart's theorem: $a(d^2+mn) = b^2m + c^2n$.

2 Problems

Don't be afraid to ask for hints, these problems are *hard*. Good luck!

Problem 1 (AHSME 1961/40). Find the minimum possible value of $\sqrt{x^2+y^2}$ if $5x+12y = 60$.

Problem 2 (AoPS Intermediate Algebra 12.75). Prove that for all real numbers a, b, x, y,

$$\sqrt{a^2+b^2} + \sqrt{x^2+y^2} \geq \sqrt{(a-x)^2 + (b-y)^2}.$$

Problem 3 (Baltic Way). Let a, b, and c be positive reals. Show that

$$\sqrt{a^2-ab+b^2} + \sqrt{b^2-bc+c^2} \geq \sqrt{a^2+ac+c^2}.$$

Problem 4 (RAMC 2023/15). For positive real numbers a, b, c,

$$\begin{aligned} a^2 - ab + b^2 &= 121 \\ a^2 - ac + c^2 &= 484 \\ b^2 + bc + c^2 &= 1089. \end{aligned}$$

Compute bc.

Problem 5 (Inspired by Angie). Over positive reals a and b, compute the maximum possible value of

$$\frac{1}{a^2b^2}\big((a+b)^2 - 2025\big)\big(2025 - (a-b)^2\big).$$

Problem 6 (Prize 2022/1)**.** Compute the value of t such that

$$\sqrt{t^2 + (t^2 - 1)^2} + \sqrt{(t - 14)^2 + (t^2 - 46)^2}.$$

Problem 7 (AIME 1991/15)**.** For positive integer n, let S_n be the minimum value of the sum

$$\sum_{k=1}^{n} \sqrt{(2k - 1)^2 + a_k^2},$$

where $a_1, a_2, \ldots, a_n$ are positive real numbers whose sum is 17. There is a unique positive integer n for which S_n is an integer. Find this n.

Problem 8 (2025 AIME I/8)**.** Let k be a real number such that the system

$$\begin{aligned} |25 + 20i - z| &= 5 \\ |z - 4 - k| &= |z - 3i - k| \end{aligned}$$

has exactly one complex solution z. Compute the sum of all possible values of k.

Problem 9 (CMIMC Algebra 2016/6)**.** For some complex number ω with $|\omega| = 2016$, there is some real $\lambda > 1$ such that ω, ω^2, and $\lambda\omega$ form an equilateral triangle in the complex plane. Find λ.

Problem 10 (AIME II 2006/15)**.** Solve over real numbers the system of equations

$$\begin{aligned} x &= \sqrt{y^2 - \frac{1}{16}} + \sqrt{z^2 - \frac{1}{16}} \\ y &= \sqrt{z^2 - \frac{1}{25}} + \sqrt{x^2 - \frac{1}{25}} \\ z &= \sqrt{x^2 - \frac{1}{36}} + \sqrt{y^2 - \frac{1}{36}}. \end{aligned}$$

Problem 11 (HMMT Algebra 2014/9)**.** Given that a, b, and c are complex numbers satisfying

$$\begin{aligned} a^2 + ab + b^2 &= 1 + i \\ b^2 + bc + c^2 &= -2 \\ c^2 + ca + a^2 &= 1, \end{aligned}$$

compute $(ab + bc + ca)^2$.

Problem 12 (AIME II 2013/15)**.** In obtuse triangle ABC with $\angle B > 90^\circ$ we have

$$\begin{aligned} \cos^2 A + \cos^2 B + 2 \sin A \sin B \cos C &= \frac{15}{8} \\ \cos^2 B + \cos^2 C + 2 \sin B \sin C \cos A &= \frac{14}{9}. \end{aligned}$$

Compute

$$\cos^2 C + \cos^2 A + 2 \sin C \sin A \cos B.$$

Problem 13 (OTIS Mock AIME I 2025/12)**.** There exists a unique tuple of positive real numbers (a, b, c, d) such that

$$(49 + ab)(a + b) = 81a + 25b$$

$$\begin{aligned}(81+bc)(b+c)&=121b+49c\\(121+cd)(c+d)&=169c+81d\\a+b+c+d&=12.\end{aligned}$$

Compute d.

Problem 14 (Math Prize 2022/18)**.** Let A be the locus of points (α, β, γ) in the $\alpha\beta\gamma$-coordinate space that satisfy the following properties:

- We have α, β, $\gamma > 0$ and $\alpha + \beta + \gamma = \pi$.
- The intersection of the three cylinders in the xyz-coordinate space given by the equations

  $$\begin{aligned}y^2+z^2&=\sin^2\alpha\\z^2+x^2&=\sin^2\beta\\x^2+y^2&=\sin^2\gamma\end{aligned}$$

  is nonempty.

Determine the area of A.

Problem 15 (HMMT Team 2022/6)**.** Let $a, b \in \mathbb{C}$. The roots of $x^4 + ax^3 + bx^2 + x$ lie on a circle in the complex plane. Prove $ab \neq 9$.